

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	Phase 3 Study of Pembrolizumab in Combination With Carboplatin/Taxane (Paclitaxel or Nab-paclitaxel) Followed by Pembrolizumab With or Without Maintenance MK-2870 in the First-line Treatment of Metastatic Squamous Non-small Cell Lung Cancer
Short Title:	Pembrolizumab With or Without Maintenance Sacituzumab Tirumotecan (Sac-TMT; MK-2870) in Metastatic Squamous Non-small Cell Lung Cancer (NSCLC) [MK-2870-023]
Protocol Number:	1029816075870010455
NCT Number:	NCT06422143
Sponsor Name:	Merck
Investigational Product:	carboplatin
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Overall survival (OS): OS is the time from randomization to death due to any cause.
Secondary Obj.:	1. Progression-Free Survival (PFS) 2. Number of Participants With ≥1 Adverse Event (AE) 3. Number of Participants Discontinuing from Study Therapy Due to AE(s)

2.2 Background & Rationale

This is a phase 3 study of pembrolizumab in combination with carboplatin/taxane (paclitaxel or nab-paclitaxel) followed by pembrolizumab with or without maintenance sacituzumab tirumotecan (sac-TMT; MK-2870) in first-line treatment of metastatic squamous non-small cell lung cancer. It is hypothesized that pembrolizumab with maintenance sacituzumab tirumotecan is superior to pembrolizumab without sacituzumab tirumotecan maintenance with respect to overall survival (OS).

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE3, Status: RECRUITING
Target N:	851

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- * Histologically or cytologically confirmed diagnosis of squamous non-small cell lung cancer (NSCLC) [Stage IV: M1a, M1b, M1c, American Joint Committee on Cancer Staging Manual, version 8]
- * Measurable disease per Response Evaluation Criteria in Solid Tumours (RECIST) 1.1 as assessed by the local site investigator/radiology

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- * Has life expectancy ≥3 months
- * Has Eastern Cooperative Oncology Group Performance Status (ECOG PS) score of 0 or 1 assessed within 7 days prior to allocation
- * Archival tumor tissue sample or newly obtained core, incisional, or excisional biopsy of a tumor lesion not previously irradiated has been provided
- * Human immunodeficiency virus (HIV)-infected participants must have well controlled HIV on antiretroviral therapy (ART)
- * Participants who are hepatitis B surface antigen (HBsAg)-positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy for at least 4 weeks and have undetectable HBV viral load before allocation
- * Participants with history of hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable at screening
- * Participants who have adverse events (AEs) due to previous anticancer therapies must have recovered to Grade ≤1 or baseline (participants with endocrine-related AEs who are adequately treated with hormone replacement are eligible).
Note: Participants with Grade =2 neuropathy are eligible
- * Has adequate organ function
- * For Maintenance only (prior to randomization): is without disease progression of their NSCLC, as determined by BICR using RECIST 1.1 after completion of study-specified Induction with an evaluable scan at Week 12 or most recent scan before randomization
- * For Maintenance only (prior to randomization): has ECOG PS of 0 or 1 as assessed at the Prerandomization Visit
- * For Maintenance only (prior to randomization): all AEs (with the exception of alopecia, Grade 2 fatigue, and Grade ≤2 endocrine-related AEs requiring treatment or hormone replacement) have recovered

4.2 Exclusion Criteria

- * Diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements
- * History of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing
- * Active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease (eg, Crohn's disease, ulcerative colitis, or chronic diarrhea)
- * Has uncontrolled, significant cardiovascular disease or cerebrovascular disease including New York Heart Association Class III or IV congestive heart failure, unstable angina, myocardial infarction, uncontrolled symptomatic arrhythmia, prolongation of QTcF interval to >480 ms, and other serious cardiovascular and cerebrovascular diseases within 6 months before study intervention
- * HIV-infected participants who have been newly diagnosed or with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease
- * Received prior systemic chemotherapy or other targeted or biological antineoplastic therapy for their metastatic NSCLC. Note: Prior treatment with chemotherapy and/or radiation as a part of neoadjuvant or adjuvant therapy or chemoradiation therapy for nonmetastatic NSCLC is allowed as long as therapy was completed at least 12 months before diagnosis of metastatic NSCLC
- * Received prior therapy with an anti-programmed cell death 1 protein (PD-1), anti-programmed cell death ligand 1 (PD-L1), or anti programmed cell death ligand 2 (PD-L2) agent, or with an agent directed to another stimulatory or coinhibitory T-cell receptor (eg, cytotoxic T lymphocyte-associated protein 4, OX-40, CD137). Note: Prior treatment with an anti-PD-1 or anti-PD-L1 agent for nonmetastatic NSCLC is allowed as long as therapy was completed at least 12

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months before diagnosis of metastatic NSCLC

- * Received prior treatment with a trophoblast cell-surface antigen 2 (TROP2)-targeted antidrug conjugate (ADC)
- * Received radiation therapy to the lung that is ≥ 30 Gray within 6 months of start of study intervention
- * Received prior radiotherapy within 2 weeks of start of study intervention, or radiation-related toxicities, requiring corticosteroids
- * Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention
- * Participants who have not adequately recovered from major surgery or have ongoing surgical complications
- * Received prior treatment with a topoisomerase I inhibitor-containing ADC
- * Is currently receiving a strong inducer/inhibitor of CYP3A4 that cannot be discontinued for the duration of the study (the required washout period before starting sac-TMT is 2 weeks)
- * Has a known additional malignancy that is progressing or has required active treatment within the past 3 years.
- * Has known central nervous system (CNS) metastases/carcinomatous meningitis
- * Severe hypersensitivity (?Grade 3) to study intervention and/or any of its excipients or to another biologic therapy
- * Active autoimmune disease that has required systemic treatment in the past 2 years (replacement therapy [eg, thyroxine, insulin, or physiologic corticosteroid] is allowed)
- * History of (noninfectious)pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease
- * Active infection requiring systemic therapy
- * History of allogeneic tissue/solid organ transplant

11. Study Identifiers & Governance

Primary ID: 1029816075870010455

Registry Number: NCT06422143

Sponsor: Merck

Study Title: Pembrolizumab With or Without Maintenance Sacituzumab Tirumotecan (Sac-TMT; MK-2870) in Metastatic Squamous Non-small Cell Lung Cancer (NSCLC) [MK-2870-023]

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____